Information Theory, Coding and Inference **2026**

Exam 1: February 2026

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Instructions: *This is a closed-book exam. You are not permitted to refer to any material or discuss the problem with anyone. Malpractice will be severely punished. Please mention your ROLL Number and name clearly in the answer sheet.*

Duration: 75 minutes

Question 1.1 (3pts). Consider a random variable supported over the set of positive integers. If the k'th moment is $\mathbb{E}[X^k] = \infty$, then what can you say about higher-order moments? More generally, if $\mathbb{E}[X^k] = \alpha$, what can you say about higher-order moments?

Question 1.2 (3+3pts). Are each of the following convex/concave/neither?

1. $f : \mathbb{R}^n \to \mathbb{R}$, with $f(x^n) = e^{\sum_{i=1}^n x_i}$
2. $f : \mathbb{R}^n \to \mathbb{R}$, with $f(x^n) = \sum_{i=1}^n e^{x_i}$

Question 1.3 (3+3+3pts). Entropy of sums: Let $Y = X + Z$, for any pair of random variables (X, Z) on $\mathcal{X} \times \mathcal{Z}$, where $\mathcal{X}, \mathcal{Z}$ are discrete subsets of $\mathbb{R}$.

- Show that $H(Y) \geqslant H(X)$ if (X, Z) are independent.
- Give an example of a joint distribution (not necessarily independent) p_{XZ} with $|\mathcal{X}| = |\mathcal{Z}| = 2$ for which $H(Y) < H(X)$
- Give an example of a joint distribution p_{XZ} with $|\mathcal{X}| = |\mathcal{Z}| = 2$ for which $H(Y) = H(X) + H(Z)$

Question 1.4 (5pts). Let X be a Poisson(α) random variable, for mean/intensity parameter $\alpha > 0$. Therefore,

$$p_X(i) = \frac{\alpha^i e^{-\alpha}}{i!}, \qquad \text{for } i = 0, 1, 2, \ldots$$

Compute $\mathbb{E}(X)$.

Question 1.5 (2+3pts). We have a bag with m red balls, m green balls and m yellow balls. Suppose that we draw $k \geqslant 2$ balls, and let the outcomes be $X_1, \ldots, X_k$.

Compute $H(X_1)$.

Which of the following results in a higher $H(X_1, \ldots, X_k)$ and why:

- The balls are drawn with replacement
- The balls are drawn without replacement

Question 1.6 (5pts). Suppose that there exists a fixed-length source code that maps $X^n \sim p_{X^n}$ to $\mathrm{ENC}(X^n) = C^{nR}$, with the property that the probability of error $P_e^{(n)} = \Pr[\hat{X}^n \neq X^n] = o(1)$.

Use this to demonstrate the existence of a variable-length lossless source code with zero probability of error, with expected codeword length $\leqslant n(R + o(1))$.

Question 1.7 (5pts). Let $X_1, X_2, \ldots$ form a first-order Markov chain. Show that $H(X$[illegible]$|X_n)$ is a nondecreasing function of n.